N1 suppression to audiovisual self-generated sensory events

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Abstract

The ability to discriminate between self-generated and externally generated events is ~~doubtlessly~~ crucial for adaptive behavior ~~comprehensive understanding of human cognition, emotional well-being, and social interaction~~. Attenuated N1 responses to self-initiated sensory events indicate a successful discrimination process based on internal prediction models. The sensory attenuation in the N1 (and P2) component has been shown across different sensory modalities (audition, vision, touch), emphasizing the modality-independence of the underlying mechanism. However, N1 sensory attenuation was only studied in isolation across different sensory modalities. Therefore, the current study investigated the N1 (and P2) sensory attenuation using self-initiated audiovisual stimuli known to be the most frequent combination of sensory events in our environment. Two EEG studies were conducted using different visual stimuli (i.e., Gabor patches vs. checkerboard reversals), each combined with tones. Each study compares the N1 sensory attenuation for unisensory stimulus presentation (only visual stimuli or auditory tones) with the multisensory presentation of audiovisual sensory events. Multisensory audiovisual stimuli elicited N1 attenuation comparable to the attenuation observed for unisensory auditory stimuli. In contrast, no N1 attenuation was observed for unisensory visual stimuli. These findings are consistent with the presence of parallel, action-dependent mechanisms operating across modalities in the present paradigm, yet they do not permit a clear distinction between existing theoretical accounts. ~~Multisensory audiovisual sensory events elicited an attenuated N1 response similar to the N1 response resulting from unisensory auditory tones. No N1 sensory attenuation was observed for unisensory visual stimuli. Multisensory audiovisual N1 attenuations were mainly driven by the auditory modality, highlighting the supremacy of the auditory modality for N1 sensory attenuations.~~

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Keywords: Sensory attenuation, Self-initiated actions, Multisensory stimuli, N1, EEG, prediction

INTRODUCTION

Differentiating self-generated from external sensory information is fundamental for adaptive perception and social interaction (Ford & Mathalon, 2012; Frith, 2005; Wolpert et al., 2003), yet its underlying mechanisms remain unclear. Neuroscientific frameworks provide a structure for understanding such complex processes (Fitch, 2014; ~~Dayan & Abbott, 2001 ;~~ Phillips, 2015; Levenstein et al., 2023). Two prominent accounts explain how the brain distinguishes self-generated from external sensory input. Internal prediction models suggest the brain uses forward and inverse models to generate modality-specific predictions, comparing expected and actual outcomes (Wolpert & Miall, 1996; Wolpert & Kawato, 1998; Wolpert et al., 2001; von Holst & Mittelstaedt, 1950; Synofzik, 2015; Synofzik et al., 2008), with sensory attenuation (SA) occurring when predictions match outcomes, explaining phenomena like the inability to tickle oneself (Blakemore et al., 1998, 1999, 2000). Predictive processing views the brain as an active, hierarchical inference system that updates predictions based on errors and SA arises from reduced processing of any well-predicted input, not just self-generated events (Friston, 2005, 2010; Friston et al., 2016; Kaiser & Schütz-Bosbach, 2018; Kiepe et al., 2021; Lange, 2009),~~(Friston, 2005, 2010, 2016; Kiepe et al., 2021; Kaiser & Schütz-Bosbach, 2018; Lange, 2009),~~ reflecting general, precision-weighted prediction. Both frameworks explain SA within a single modality, but their implications for multisensory contexts are unclear. Integrating expectations across channels may enhance precision (de Lange et al., 2018; Talsma, 2015), yet whether this affects SA, and how the two frameworks differ, remains unknown. Multisensory integration has been shown to enhance sensory processing, with converging inputs leading to amplified neural responses compared to unisensory stimulation (Giard & Peronnet, 1999; Meredith, 2002). Such enhancement effects can emerge at early stages of processing and reflect integrative mechanisms that combine information across modalities.

N1 sensory attenuation in audition and vision

SA is typically indexed by N1 (and less consistently, P2) ERPs, reflecting neural differences between self-generated and external events (for review, Bendixen et al., 2012; Horváth, 2015; Hughes et al., 2013b; Lange, 2013). N1 reflects early sensory detection and discrimination (Mariano et al., 2024; Näätänen & Picton, 1987; Vogel & Luck, 2000; Scanlon et al., 2019~~8~~), while P2 relates to evaluation, classification, and attention (Crowley & Colrain, 2004; Han et al., 2021; Shahin et al., 2005; Korka et al., 2022; Bolt & Loehr, 2023; Ghio et al., 2021; Pinheiro et al., 2019). Auditory N1 attenuation is robust across tasks and manipulations (e.g. Bäß et al., 2008; Chen et al., 2012; Ford et al., 2007; Harrison et al., 2021; Heinks-Maldonado et al., 2005; Hughes et al., 2013a; Klaffehn et al., 2019; Knolle et al., 2013b, 2019; Kühn et al., 2011; Lange, 2011; Paraskevoudi & SanMiguel, 2023; SanMiguel et al., 2013; Seidel et al., 2021; Timm et al., 2016~~Bäß et al., 2008; Chen et al., 2012; Lange, 2011; Harrison et al., 2021; Hughes et al., 2013a; Ford et al., 2007; Klaffehn et al., 2019; Knolle et al., 2012, 2013b;~~

~~SanMiguel et al., 2013; Kühn et al., 2011; Seidel et al., 2021; Timm et al., 2016; Heinks-Maldonado et al., 2005~~). Visual findings are mixed: some report N1 or P2 attenuation for self-initiated visual events (Gentsch et al., 2012; Gentsch & Schütz-Bosbach, 2011; Mifsud et al. 2018; Ody et al., 2023; Schafer & Marcus, 1973), while others show N1 enhancement (Balla et al., 2020; Mifsud et al., 2016) or no effect (Balla et al., 2024; Csifcsák et al., 2019; Hughes & Waszak, 2011), likely due to methodological differences (e.g., stimulus type or presentation). Few studies comparing auditory and visual attenuation within the same subjects found either SA in both modalities (Schafer & Marcus, 1973) or selective effects: Mifsud et al. (2016) observed auditory N1/P2 attenuation but enhanced visual N145, whereas Ody et al. (2023) found visual N1/P2 attenuation with enhanced auditory N1 and no auditory P2 effect.

The present study

Previous research comparing auditory and visual SA has yielded mixed results, raising the question of whether these differences reflect distinct modality-specific mechanisms or variations in predictive precision. All prior EEG studies focused on unisensory events and did not test multisensory predictions following self-initiated actions. We examined SA after self-generated actions in auditory, visual, and audiovisual ~~(multisensory)~~ conditions. Presenting audiovisual stimuli allowed us to test whether integrating signals enhances prediction precision and SA, as proposed by predictive processing. Internal prediction models suggest distinct, parallel predictions for each modality (Wolpert & Kawato, 1998). We hypothesized that (i) self-initiated audiovisual events would elicit the largest N1 attenuation, and (ii) both auditory and visual self-initiated events would show N1 attenuation without a predicted directionality of modality effects. These hypotheses were tested in two EEG experiments using visual pattern reversals or Gabor patches.

EXPERIMENT 1

A procedure inspired by the one used in the study by Mifsud et al. (2016) was implemented. Here, three active and three passive conditions with sensory events were used. In the active conditions, self-initiated button presses either triggered a sound (unisensory auditory), or elicited a pattern reversal (unisensory visual), or elicited a pattern reversal combined with a sound (multisensory audiovisual). The exact sequence of sensory events in either modality was online recorded and played back as externally initiated sensory events in the passive conditions. Thus, we used the original approach of collecting the actual sequence of sensory events created by the participant’s button presses (cf. Bäß et al., 2008; Ford et al., 2007; Martikainen et al., 2005; Schafer & Marcus, 1973), ensuring the timing and sequence of sensory events are consistent across conditions, rather than using randomly generated sequences of sensory events in the passive conditions (cf. Mifsud et al., 2016; Ody et al., 2023). Consequently, the active conditions always preceded the passive ones. In addition, a motor condition was used to assess the motor activity linked to the button presses without the presence of a sensory event as a control condition.

Methods

Participants

Twenty-eight students were recruited for the experiment. Data from three participants were excluded from further analysis due to excessive artifacts. In addition, two data sets were not recorded for technical reasons. Consequently, the final sample consisted of 23 participants (17 females and 6 males, M_{age} = 22.73 years, 4.78 *SD*, 17 to 40 years old). G*Power software version 3.1 (Faul et al., 2007) for an ANOVA was used to calculate the required sample size for achieving 80% power in detecting a 0.25 effect size at an alpha level of .05, which showed that a sample size of N= 19 was sufficient. 22 participants were right-handed (mean handedness quotient: 81.96, 13.88 *SD*) and one was ambidextrous according to the Edinburgh Handedness Inventory (Oldfield, 1971).

All participants reported no neurological disorder and had normal or corrected-to-normal vision and normal hearing. All participants studied at the University of Hildesheim and received partial course credit or monetary compensation of 30€ for their participation. All participants gave written informed consent and were treated according to the Declaration of Helsinki. The research study was approved by the local ethics committee (Fachbereich 1) of the University of Hildesheim.

Design

The current study employed a 3 (Sensory Modality: auditory, visual, audiovisual) x 2 (Task: self-initiated, externally-initiated) within-subjects factorial design. In addition, a motor-only self-initiated task was utilized to control for the motor activity when pressing a button. N1 and P2 amplitudes were measured as dependent variables.

Stimuli

Auditory task

A white central fixation cross (1.06 cm in both width and height, line thickness: 0.19 cm) was consistently displayed. After each self-initiated button press, the auditory stimulus was presented (1000 Hz sinus wave tone, 50 ms duration with a ramp of ± 10 ms), played at an approximate sound pressure level (SPL) of 85 decibels presented binaurally through headphones (Sennheiser HD 280 pro 64). The timing of the sound sequence was stored online and played back as externally initiated sounds.

Visual task

A checkerboard (8.77 degrees of visual angle, with a luminance of 120 lux measured with a Gossen, Mavolux 5032c Base), consisting of a 3×3 grid with alternating black (RGB: 0, 0, 0) and white (RGB: 255, 255, 255) squares, was shown as a visual stimulus. At the center of the checkerboard, a fixation dot was presented in gray (RGB values: 134,130,125). The checkerboard always started with two black squares (i.e. left and right) and one white square (middle) in the top row; one black square (i.e. middle) and two white squares (i.e. left and right) in the middle row; and again, two black squares (i.e. left and right) and one white square (middle) in the bottom row.

After a self-initiated button press, the colors of the checkerboard switched, i.e., white squares became black and the black squares became white. After 1,500 ms, the pattern reversed again to the initial one. The timing of the pattern reversals was stored online and played back as externally initiated pattern reversals.

Audiovisual task

A checkerboard (for details, see visual task) combined with a tone (for details, see auditory task) was used which was presented throughout the whole block. After a self-initiated button press, the colors of the checkerboard switched and the tone was simultaneously presented. After 1,500 ms, the pattern changed back to the initial one.

Procedure

Participants were seated in a comfortable chair inside a sound-attenuated cabin with their right hand resting on a keyboard. They were positioned in front of a Miro 17-inch CRT monitor at a viewing distance of approximately 150 cm outside of the cabin. Participants received instructions to maintain a state of minimal movement and to reduce eye blinking as much as possible throughout the experiment.

The collection of data and the display of stimuli were conducted utilizing the Presentation® software (Version 23.0, Neurobehavioral Systems, Inc., Berkeley, CA, www.neurobs.com). All instructions were given on the computer screen. All visual stimuli were presented at the screen's center and against a gray background (RGB values: 125,125,125).

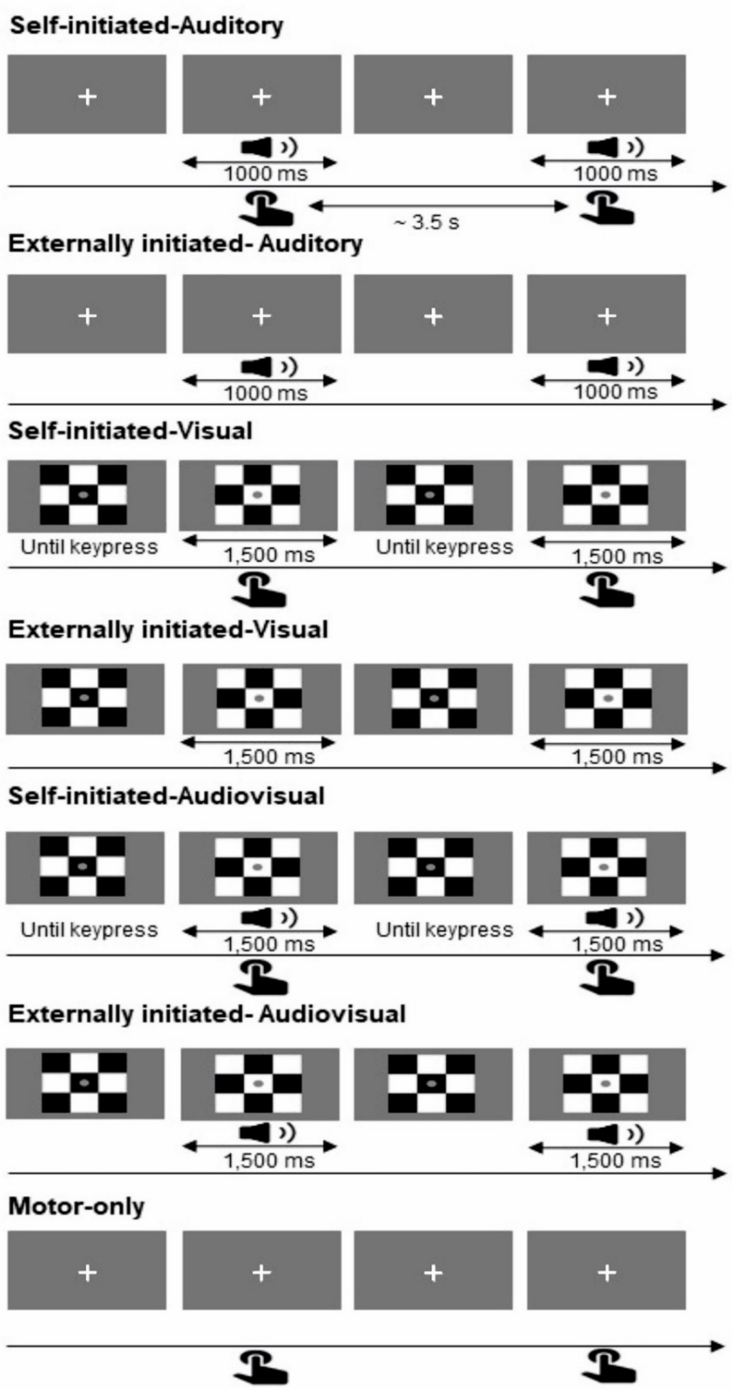
Participants performed through three self-initiated and three externally initiated passive tasks and one motor-only self-initiated task (see Figure 1). Participants were instructed to press the button once every 3.5 seconds during the self-initiated active tasks. Keypresses were required by pressing the F key with the right index finger. A short training (40 trials) was run before the start of the experimental conditions, providing feedback about the temporal distance to the previous trial (correct, too fast (< 2500 ms), or too slow (> 4500 ms)) after each trial. For the main blocks, feedback was given at the end of each block (average distance between button presses, frequency of too fast or too slow responses). For the externally initiated passive tasks, participants were instructed to listen to or view the stimuli without pressing any keys. In the motor-only task, keypresses were performed without any stimuli serving as a control for the motor activity linked with the self-initiated button press.

Each experimental task consisted of four blocks of 40 trials each, with each block repeated four times. This resulted in 160 trials per condition and a total of 1120 trials over the course of the experiment. Blocks were presented in a quasi-random order, with the constraint that, for each sensory modality, self-initiated (active) blocks always preceded externally initiated (passive) blocks. ~~constrained by having externally initiated blocks always precede self-initiated blocks for each sensory modality.~~ The timing of the self-initiated stimuli was recorded and subsequently used to generate the sequence of events in the corresponding externally initiated blocks. Participants took self-paced breaks between the different blocks.

Upon completion of the experiment, participants received compensation and underwent a debriefing process.

Figure 1

Experimental Conditions and Stimulus Presentation in Experiment 1



Note. Schematic representation of the experimental conditions in Experiment 1, showing the different task types and sensory modalities (auditory, visual, and audiovisual). A motor-only condition was included as a control task without any stimulus presentation. The figure shows the timing and order of stimulus presentation for each modality and task type.

EEG recording and analysis

EEG data were continuously recorded with a BioSemi Active Two system using sintered Ag-AgCl electrodes in an electrode cap at 32 scalp sites (Fp1, Fp2, AF3, AF4, F7, F8, F3, F4, FC1, FC2, FC5, FC6, T7, T8, C3, C4, CP1, CP2, CP5, CP6, P7, P8, P3, P4, Pz, PO3, PO4, O1, O2, Oz, Fz, Cz), positioned according to the 10-20 system, at a sampling rate of 2048 Hz. During recording, the Common Mode Sense (CMS) and Driven Right Leg (DRL) were used as a reference and the driven right leg (DRL) as the ground electrode respectively. The bipolar vertical electrooculogram (EOG) was derived from electrodes placed both above and below the right eye (VEOG) and the horizontal EOG was derived from electrodes on the right and left outer canthi (HEOG). Further external electrodes were placed on the left and right mastoids as well as on the nose and the right earlobe. All electrode impedances were maintained below 10 kΩ.

The data were processed offline using *BrainVision Analyzer* (*BrainVision Analyzer, Version 2.2.0, Brain Products GmbH, Gilching, Germany)*. EEG data were down-sampled to a 512 Hz sampling rate. All channels were re-referenced to the average of the two mastoid electrodes. Data were band-pass filtered from 0.3 to 25 Hz using a Butterworth filter and separated into 600-ms epochs, including a 100-ms prestimulus baseline. ~~An additional 50 Hz notch filter was employed.~~ Key press intervals shorter than 2500 ms or longer than 4500 ms were treated as errors, and respective trials were removed from further analysis. Data were baseline corrected from -100 to 0 ms. All individual trials with EEG or EOG activity exceeding 100 μV on any channel were excluded from further processing. Individual trials for each condition were averaged for each participant. Grand average ERP waveforms were computed from the individual average ERPs

In order to control for the motor activity of a button press, motor-only waveforms were subtracted from the self-initiated active waveforms for all three tasks (i.e., motor-auditory, motor-vision, motor-audiovisual) to produce difference waveforms as is commonly done (Bäß et al., 2008; Ford et al., 2007; Kiepe et al., 2021; Martikainen et al., 2005). All analyses were done using the self-initiated active tasks corrected for motor activity.

Time windows for the statistical analysis were set based on the difference waves between the externally initiated passive tasks (i.e., auditory-only, vision-only, audiovisual-only) and the self-initiated active tasks (i.e., motor-auditory, motor-vision, and motor-audiovisual, all corrected for motor-only activity). These difference waves showed two effects: one in the N1 time window and one in the P2 time window. Time windows were set around the maxima of all three difference waves for the N1 component (153-193 ms) and the P2 (242-282 ms)[1].

For all statistical analyses, individual ERP component amplitudes were measured as the mean of the voltage in each condition for each time window.

Statistical analysis

Repeated measurement analysis of variance (ANOVA) with the within-subject factors Sensory Modality (auditory, visual, and audiovisual) and Task (self-initiated and externally initiated) was calculated separately for the N1 and P2 components at the Cz electrode following visual inspection of the topography maps and grand averages. Attenuation effects were calculated as the difference between the externally initiated stimuli and the self-initiated stimuli (corrected for the motor activity).

[1] To ensure that the results were not dependent on the choice of the time window and further validate this approach, additional analyses were conducted using time windows centered on peak latencies identified in the ERPs across the conditions. These analyses yielded the same pattern of results (see Appendix). ~~To assess the robustness of this approach, additional analyses were conducted using time windows centered on peak latencies identified in the grand-average ERP across the original (non-differenced) conditions, resulting in intervals of 115–155 ms for the N1 and 202–242 ms for the P2.~~

When applicable, the *p*-value was corrected for multiple tests. If necessary, Greenhouse-Geisser's corrected *p*-values are reported along with the corrected degrees of freedom. The statistical analyses were executed utilizing IBM SPSS Statistics for Windows, Version 28 (IBM Corp., Armonk, NY, USA).

Results

N1 time window

The omnibus ANOVA[2] with the factors Sensory Modality and Task revealed a significant main effect of Sensory Modality, $F(2, 44) = 21.77$, $p < .001$, $\eta_p^2 = .497$, showing that the largest N1 amplitudes were elicited by the audiovisual modality (M = -4.39 μV, SD = 2.94), which were significantly larger than those observed for the auditory modality (M = -2.70 μV, SD = 2.55; $t(22) = 3.07$, $p = .006$) and the visual modality (M = -0.45 μV, SD = 1.87; $t(22) = 6.42$, $p < .001$). Additionally, the auditory modality evoked significantly larger N1 amplitudes than the visual modality ($t(22) = 3.57$, $p = .002$). (see Figure 2 and Figure 3). A main effect of Task, $F(1, 22) = 17.56$, $p < .001$, $\eta_p^2 = .444$, was also observed, pointing towards smaller N1 amplitude for the self-initiated (M = -1.67 μV, SD = 2.19) compared to the externally initiated tasks (M = -3.36 μV, SD = 2.00). Importantly, a significant interaction between Sensory Modality and Task was observed, $F(2, 44) = 4.58$, $p = .016$, $\eta_p^2 = .172$.

To address the expectations regarding differences in amplitudes between self- versus externally initiated tasks for each sensory modality, post-hoc t-tests were conducted, showing that self-initiated auditory stimuli (M = -1.59 μV, SD = 2.76) elicited significantly smaller N1 responses than externally initiated auditory stimuli (M = -3.80 μV, SD = 2.81; $t(22) = 4.72$, $p < .001$). Likewise, self-initiated audiovisual stimuli (M = -3.24 μV, SD = 2.86) produced significantly smaller N1 responses

[2] In line with the analyses of Mifsud et al. (2016), where visual blocks were re-referenced to the Fz electrode, and analyses conducted at Oz, additional analyses were also performed for the visual conditions (see Figure 6 in the Appendix for ERPs), which yielded no significant differences between self- and externally initiated stimuli.

compared to externally initiated audiovisual stimuli (*M* = -5.55 µV, *SD* = 3.59; *t*(22) = 4.03, *p* < .001). For the visual modality, the corresponding comparison showed no significant difference between self-initiated and externally initiated stimuli, *t*(22) = 0.92, *p* = .367.

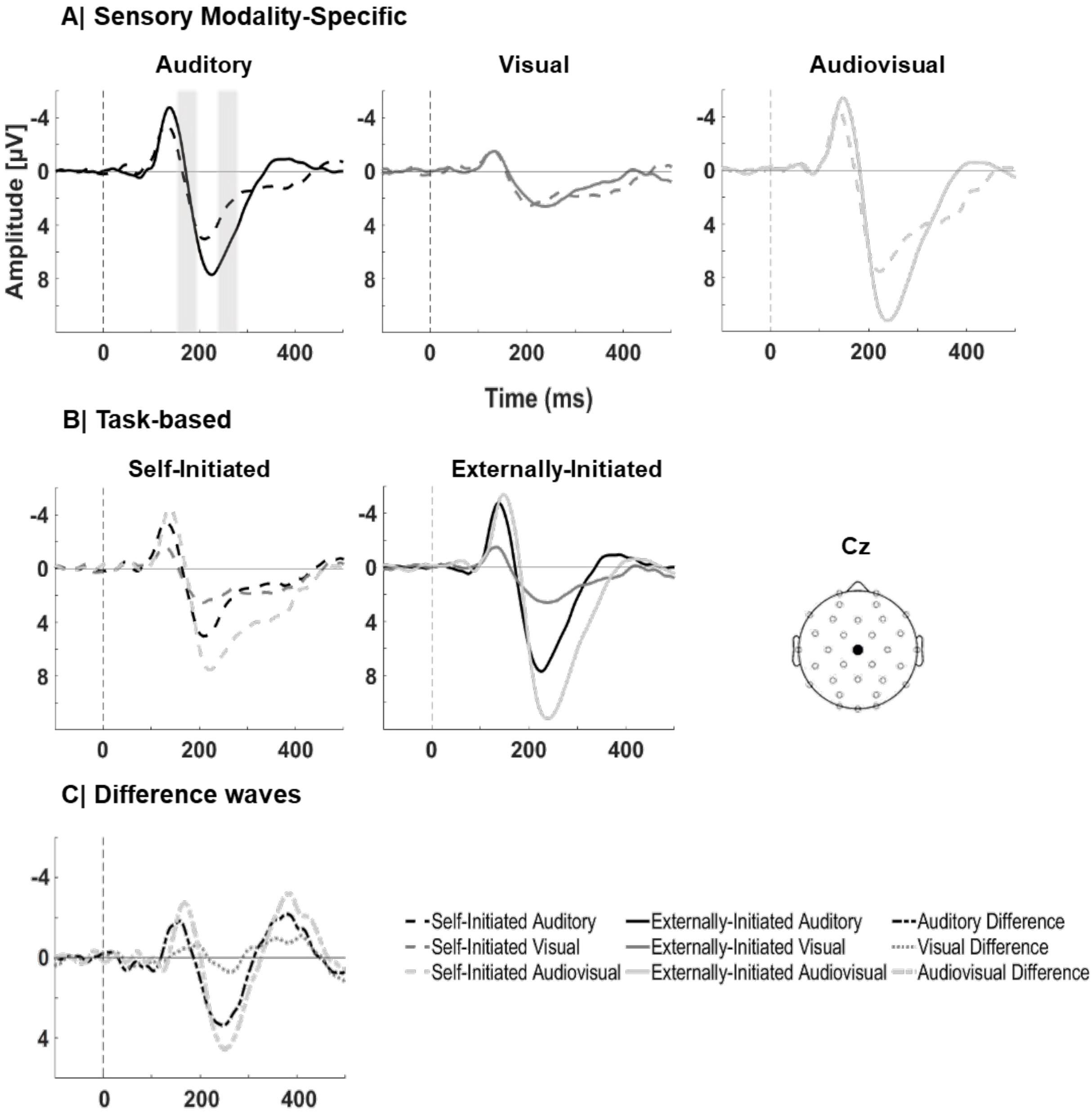
To address expectations regarding differences in amplitudes between self-initiated audiovisual, auditory, and visual modalities, post-hoc t-tests for self-initiated active tasks were conducted, pointing to a smaller N1 amplitude for the visual (*M* = -0.17 µV, *SD* = 2.98) compared to the audiovisual stimuli (*M* = -3.24 µV, *SD* = 2.86), *t*(22) = 4.74, *p* < .001. Further, the N1 amplitude was significantly smaller for the auditory (*M* = -1.59 µV, *SD* = 2.76) compared to the audiovisual stimuli, *t*(22) = 2.72, *p* = .013. No significant difference was found between the auditory and visual stimuli, *t*(22) = -1.91, *p* = .070. For externally initiated passive tasks, post hoc t-tests indicated that the largest N1 amplitude was elicited by the audiovisual stimuli (*M* = -5.55 µV, *SD* = 3.59), which was significantly larger than for the auditory (*M* = -3.80 µV, *SD* = 2.81; *t*(22) = 2.91, *p* = .008) and visual stimuli (*M* = -0.73 µV, *SD* = 1.50; *t*(22) = 6.11, *p* < .001). Additionally, auditory stimuli elicited significantly larger N1 amplitudes than visual stimuli, ~~(~~*t*(22) = 4.50, *p* < .001~~)~~.

To further examine whether there are differences in N1 attenuation effects based on sensory modality, the attenuation effects as the difference between externally initiated and self-initiated tasks for each modality were compared by performing post-hoc tests, showing that differences in N1 amplitudes across the auditory, visual, and audiovisual modalities did not reach statistical significance and were therefore only evident at a descriptive level, *ts*(22) ≤ 2.41, *ps* ≥ .025, Bonferroni-corrected.

To rule out potential confounds related to the sequence of block presentation, the omnibus ANOVA was repeated including sensory modality order as a between-subject factor. This analysis revealed no significant main effect of sensory modality order, $F(2, 20) = 1.48$, $p = .252$, $\eta p^2 = .129$. While there were no significant interactions between Order and either Modality or Task (all *p*s ≥ .30), a significant Modality × Task × Order interaction was observed, $F(4, 40) = 5.86$, $p < .001$, $\eta p^2 = .370$.

Figure 2

Grand average ERPs per sensory modality elicited in self-initiated and externally initiated conditions in Experiment 1



Note. Grand average waveforms at the Cz electrode for (A) auditory, visual, and audiovisual modalities, plotted separately for self-initiated (dashed lines) and externally initiated (solid lines) stimuli. (B) Self-initiated and externally initiated tasks are displayed for each sensory modality, with self-initiated conditions represented by dashed lines and externally initiated conditions by solid lines. (C) Difference waves, computed as externally minus self-initiated conditions, are shown for each modality. Shaded gray bars highlight the analyzed time windows for the N1 and P2 components.

P2 time window

The omnibus ANOVA showed a main effect of Sensory Modality, $F(2, 44) = 54.81$, $p < .001$, $\eta_p^2 = .714$, highlighting significant differences between all modalities. The largest P2 amplitudes were elicited by the audiovisual modality ($M = 9.42$ μV, $SD = 3.53$), which were significantly larger than those observed for the auditory modality ($M = 5.29$ μV, $SD = 2.56$; $t(22) = -7.91$, $p < .001$) and the visual modality ($M = 2.82$ μV, $SD = 2.20$; $t(22) = -9.58$, $p < .001$). Additionally, the auditory modality evoked significantly larger ~~N1~~ P2 amplitudes than the visual modality ($t(22) = 3.60$, $p = .002$). The ANOVA also showed a main effect of Task, $F(1, 22) = 18.09$, $p < .001$, $\eta_p^2 = .451$, indicating larger P2 amplitudes for externally initiated stimuli ($M = 7.22$ μV, $SD = 3.19$) in comparison to self-initiated ones ($M = 4.46$ μV, $SD = 2.08$). Most importantly, the interaction between Sensory Modality and Task was significant, $F(2, 44) = 13.29$, $p < .001$, $\eta_p^2 = .377$.

Follow-up tests were calculated for each modality to compare self- versus externally initiated stimuli. For the auditory modality, P2 responses were significantly smaller for self-initiated stimuli ($M = 3.63$ μV, $SD = 2.27$) compared to externally initiated stimuli ($M = 6.94$ μV, $SD = 3.95$; $t(22) = -4.04$, $p < .001$). Similarly, for the audiovisual modality, self-initiated stimuli ($M = 7.09$ μV, $SD = 2.85$) elicited significantly smaller P2 responses compared to externally initiated stimuli ($M = 11.75$ μV, $SD = 5.13$; $t(22) = -5.13$, $p < .001$). For the visual modality, there was no significant difference between the self-compared to externally initiated stimuli, $t(22) = -0.43$, $p = .673$.

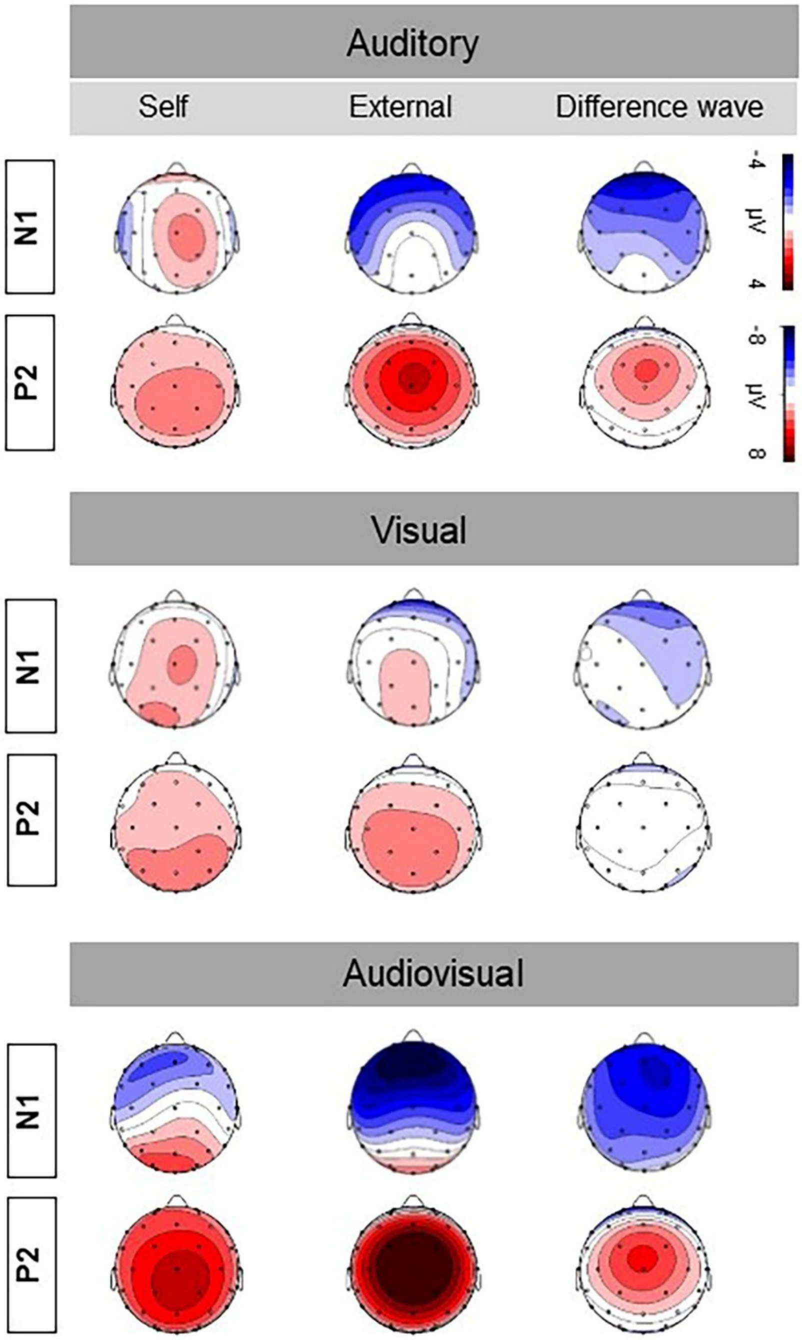
To address expectations regarding differences in amplitudes between self-initiated audiovisual, auditory and visual modalities, post-hoc t-tests for self-initiated active tasks were conducted, pointing to larger P2 amplitudes for the audiovisual stimuli (M = 7.09 μV, SD = 2.85) compared to both the auditory stimuli (M = 3.63 μV, SD = 2.27), $t(22) = -6.73$, $p < .001$, and the visual modality (M = 2.66 μV, SD = 3.35), $t(22) = -5.83$, $p < .001$. No differences in the P2 amplitudes were found between auditory and visual stimuli, $t(22) = 1.20$, $p = .242$. For externally initiated passive tasks, post hoc paired t-tests revealed that P2 amplitudes were significantly higher for audiovisual (M = 11.75, SD = 5.13) compared to both auditory (M = 6.94, SD = 3.95; $t(22) = -6.73$, $p < .001$) and visual stimuli (M = 2.97, SD = 2.09; $t(22) = -9.49$, $p < .001$), and that P2 amplitudes were also higher for auditory compared to visual stimuli, $t(22) = 4.75$, $p < .001$.

~~Lastly, post~~Post-hoc t-tests with the attenuation effects, i.e. the difference waves between externally initiated and self-initiated stimuli for each modality, showed larger P2 attenuation effect for the audiovisual (M = 5.37 μV, SD = 4.23) compared to visual modalities (M = 1.41 μV, SD = 3.76), $t(22) = -3.96$, $p < .001$. Similarly, the attenuation effect was greater for the auditory (M = 4.14 μV, SD = 3.23) compared to the visual modality, $t(22) = 3.43$, $p = .002$. No significant difference in the P2 amplitudes was found between the auditory and audiovisual modalities, $t(22) = -1.90$, $p = .071$.

To rule out potential confounds related to the sequence of block presentation, the omnibus ANOVA was repeated including sensory modality order as a between-subject factor. This analysis revealed no significant main effect of sensory modality order, $F(2, 20) = 2.21$, $p = .136$, $\eta_p^2 = .181$. A significant Modality × Order interaction was observed, $F(4, 40) = 3.28$, $p = .020$, $\eta_p^2 = .247$, whereas no other interactions involving Order were significant (all ps ≥ .092).

Figure 3

Topographic Maps of N1 and P2 Components in Experiment 1



Note. Topographic maps show the N1 and P2 components for each sensory modality. Scalp distributions are shown for the N1 and P2 time windows for self-initiated, externally initiated, and difference wave conditions.

Discussion

Experiment 1 showed attenuated N1 and P2 responses for self-initiated unisensory auditory and multisensory audiovisual actions compared to externally initiated ones, but not for unisensory visual events. This partially aligns with prior N1 attenuation literature: attenuation for unisensory auditory events is well documented (e.g., Baess et al., 2011; Heinks-Maldonado et al., 2005; Klaffehn et al., 2019; Mifsud et al., 2016; SanMiguel et al., 2013). Here, we showed that this also occurs for multisensory audiovisual events, which descriptively elicited larger N1 and P2 responses than auditory events alone. No differences were observed between self- versus externally initiated visual events. Previous studies reported either enhancement (Balla et al., 2020; Mifsud et al., 2016) or attenuation (Gentsch & Schütz-Bosbach, 2011; Ody et al., 2023; Schafer & Marcus, 1973) for visual events; in our study, no N1 attenuation was evident in the visual modality, although auditory and audiovisual events within the same sample clearly showed attenuated N1 and P2 responses. To confirm these findings, Experiment 2 used onset visual stimuli instead of pattern reversals and matched stimulus presentation across modalities.

EXPERIMENT 2

Given the diverse results on the N1 attenuation with visual stimuli (e.g. Csifcsák et al., 2019; Gentsch & Schütz-Bosbach, 2011; Mifsud et al., 2016, 2018; Ody et al., 2023), Experiment 2 was conducted to replicate Experiment 1 with different onset visual stimuli, namely Gabor patches. The Gabor patches occurred after the keypress, so apart from the fixation cross, no visual stimuli were shown before or during the keypress. Consequently, the presentation of the visual stimuli closely follows that of the auditory stimuli, i.e., presenting a sensory event after a key press. Therefore, essentially, Experiment 2 was identical to Experiment 1 except for the visual stimulus displayed in the visual and audiovisual conditions.

Methods

Participants

Thirty-eight students were recruited for the experiment. Data from eight participants were excluded from further analysis due to excessive artifacts. Consequently, the final sample consisted of 30 participants (22 females and 8 males, Mage = 23.16 years, 4.68 SD, 19 to 39 years old). G*Power software version 3.1 (Faul et al., 2007) for an ANOVA with a repeated measure, within factors was used to calculate the required sample size for achieving 80% power in detecting a 0.25 effect size at an alpha level of .05, which showed that the sample size was sufficient. All participants were right-handed, except for two left-handed, and two were ambidextrous (mean handedness quotient: 70.17, 51.72 *SD*) according to the Edinburgh Handedness Inventory (Oldfield, 1971).

All participants reported no neurological disorder and had normal or corrected-to-normal vision and normal hearing. All participants studied at the University of Hildesheim and received partial course credit or monetary compensation of 30 euros for their participation. All participants gave written informed consent and were treated according to the Declaration of Helsinki. The research study was approved by the local ethics committee (Fachbereich 1) of the University of Hildesheim. The study was preregistered at the Open Science Framework (https://osf.io/6pj7n).

Procedure and task

Experiment 2 was identical to Experiment 1, with the following exception. Instead of the checkerboard pattern reversal used in Experiment 1, the visual stimulus consisted of the onset of a vertically oriented Gabor patch (luminance: 23.5 lux; visual angle: 3.12°). The stimulus was presented as a centered 8.3 × 8.3 cm square and comprised a sinusoidal grating modulated by a Gaussian envelope (SD = 50). The spatial frequency of the grating was 0.02 cycles/pixel. The patch was presented at high contrast (Michelson contrast ≈ 1.0), with luminance values ranging from black (RGB: 0, 0, 0) to white (RGB: 255, 255, 255) on a gray background (RGB: 128, 128, 128). The Gabor patch was generated using an online Gabor stimulus generator (available at https://www.cogsci.nl/gabor-generator). ~~Experiment 2 was the same as Experiment 1, with the following exceptions. First, the usage of the onset Gabor patch with a luminance of 23.5 lux at a visual angle of 3.12 degrees, characterized by a spectrum of gray tones ranging from the darkest gray (RGB values: 15, 15, 15) to near white (RGB values: 250, 250, 250), as the visual stimulus onset instead of a checkerboard reversal pattern. The patch was presented as a 8.3 x 8.3 cm square, centered on the screen.~~ Each block began with a fixation cross, which remained visible in all conditions until the Gabor patch appeared on the keypress. Each trial lasted 1000 ms. Second, the study was carried out within an electromagnetically shielded cabin with dimmed lighting (Desone E: Box, Desone Modulare Akustik, Berlin, Germany). Participants were comfortably seated in a chair positioned in front of a Dell 23.8-inch LCD monitor placed outside of the cabin while maintaining a viewing distance of approximately 145 cm. Third, the down arrow key on a standard QWERTZ keyboard was the response button.

EEG recording and analysis

Data processing and statistical analysis were identical to Experiment 1. All individual trials with EEG activity exceeding 100 µV on any channel were excluded from further processing. Time windows for the N1 and P2 components were assessed based on the maxima of the difference waves for the three modalities in the grand average waveforms. The time window for the N1 component was 166-206 ms, and the time window for the P2 component was 248-288 ms[3]. Mean amplitudes were extracted based on individual averages.

Results

N1 time window

Group average ERPs for the stimuli in the self-initiated and externally initiated conditions are presented in Figure 4. The omnibus ANOVA[4] revealed a main effect of Sensory Modality, $F(2, 58) = 35.18$, $p < .001$, $\eta_p^2 = .548$, pointing to significant differences in the N1 component between all sensory modalities (see Figure 5). Specifically, the largest N1 amplitude was elicited by the audiovisual modality (M = -4.73 µV, SD = 2.20), which was significantly larger than the amplitudes observed for the auditory modality (M = -3.45 µV, SD = 2.00; $t(29) = 4.68$, $p < .001$) and the visual modality (M = -1.77 µV, SD = 1.11; $t(29) = 8.02$, $p < .001$). Moreover, the auditory modality elicited significantly larger N1 amplitudes than the visual modality ($t(29)$ = 4.14, $p < .001$). Additionally, a main effect of Task, $F(1, 29) = 7.57$, $p = .010$, η_p^2 = .207 was detected showing larger N1 amplitudes for the externally initiated (M = -3.62 µV, SD = 1.55) compared to self-initiated tasks (M = -3.01 µV, SD = 1.59). Of particular interest was the interaction between Sensory Modality and Task, $F(2, 58)$ = 8.71, $p < .001$, η_p^2= .231.

To address the expectations regarding differences in N1 amplitudes between self- versus externally initiated tasks for each sensory modality, post hoc t-tests showed

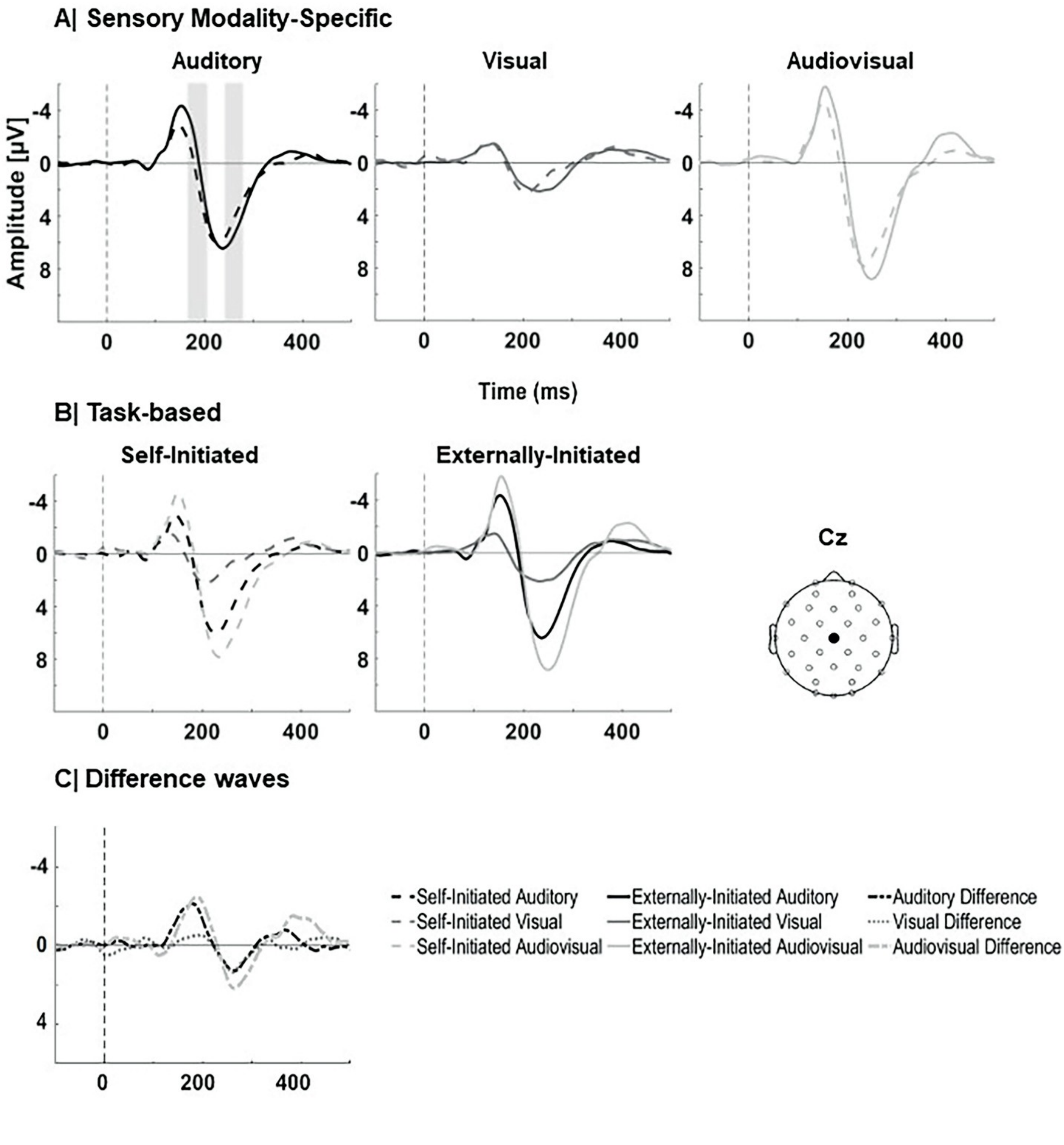
[3] --
[4] The same additional analyses as in Experiment 1 (re-referencing to Fz and analyzing at Oz) could not be conducted because no clear ERP components of interest were detectable at Oz (see Figure 7 in the Appendix).

that N1 amplitudes for self-initiated auditory stimuli (*M* = -2.83 µV, *SD* = 2.47) were significantly smaller than for externally initiated auditory stimuli (*M* = -4.06 µV, *SD* = 1.85; $t(29) = 3.85$, $p < .001$). Similarly, self-initiated audiovisual stimuli (*M* = -4.26 µV, *SD* = 2.19) elicited significantly smaller N1 amplitudes than externally initiated audiovisual stimuli (*M* = -5.20 µV, *SD* = 2.64; $t(29) = 2.56$, $p = .016$). Importantly, no differences in the N1 component were observed between self-initiated and externally initiated stimuli in the visual modality, $t(29) = -1.33$, $p = .193$.

To address expectations regarding differences in amplitudes between self-initiated audiovisual, auditory, and visual modalities, post-hoc t-tests for self-initiated active tasks were conducted. Larger N1 responses for self-initiated sensory events were observed for audiovisual stimuli (*M* = -4.26 µV, *SD* = 2.19), compared to both auditory (*M* = -2.83 µV, *SD* = 2.47; $t(29) = 4.01$, $p < .001$) and visual stimuli (*M* = -1.95 µV, *SD* = 1.63; $t(29) = 5.62$, $p < .001$). No significant differences were found between visual and auditory stimuli, $t(29) = -1.62$, $p = .116$). For externally initiated passive tasks, post hoc t-tests indicated that the largest N1 responses were elicited by the audiovisual stimuli (*M* = -5.20 µV, *SD* = 2.64), which were significantly larger than those for auditory (*M* = -4.06 µV, *SD* = 1.85; $t(29) = 3.41$, $p = .002$) and visual stimuli (*M* = -1.59 µV, *SD* = 0.92; $t(29) = 8.47$, $p < .001$). Furthermore, auditory N1 responses were significantly larger than visual N1 responses, $t(29) = 7.43$, $p < .001$. To further examine the differences in N1 attenuation effects across sensory - modalities, the difference waves were compared by performing post-hoc tests showing a larger N1 attenuation effect for the audiovisual sensory events (*M* = -2.33 µV, *SD* = 2.15) compared to the visual ones (*M* = -1.07 µV, *SD* = 2.09), $t(29) = 2.67$, $p = .012$. The attenuation effects did not differ between auditory modality and the audiovisual one nor between auditory and visual one, $ts(29) \leq -2.37$, $ps \geq .025$, Bonferroni-corrected. To rule out potential confounds related to the sequence of block presentation, the omnibus ANOVA was repeated including sensory modality order as a between-subject factor. This analysis revealed no significant main effect of sensory modality order, $F(2, 27) = 0.93$, $p = .407$, $\eta p^2 = .064$, and no significant interactions involving Order (all $ps \geq .55$), indicating that the observed effects were not influenced by block order.

Figure 4

Grand average ERPs per sensory modality elicited in self-initiated and externally initiated conditions in Experiment 2



Note. Grand average waveforms at the Cz electrode for (A) auditory, visual, and audiovisual modalities, plotted separately for self-initiated (dashed lines) and externally initiated (solid lines) stimuli. (B) Self-initiated and externally initiated tasks are displayed for each sensory modality, with self-initiated conditions represented by dashed lines and externally initiated conditions by solid lines. (C) Difference waves, computed as externally minus self-initiated conditions, are shown for each modality. Shaded gray bars highlight the analyzed time windows for the N1 and P2 components.

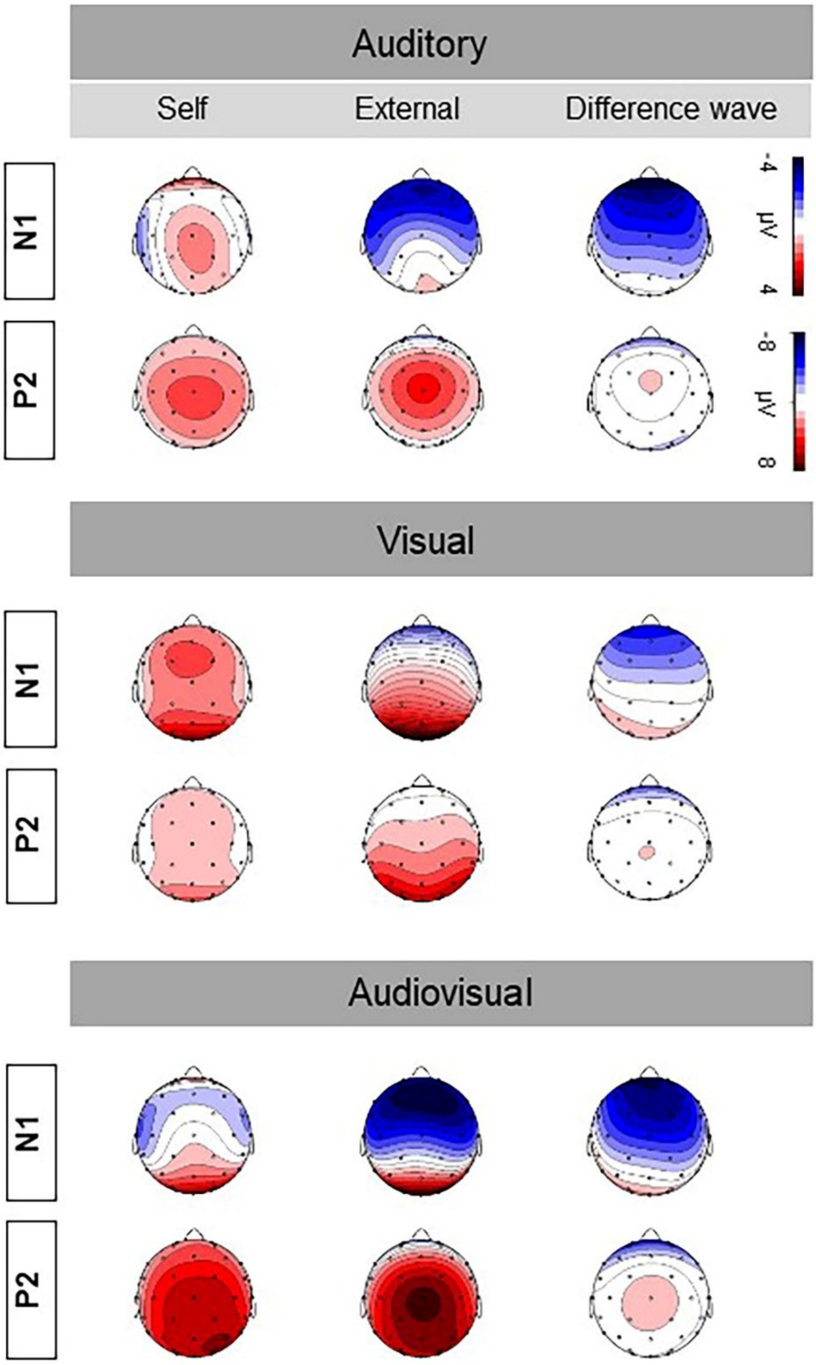
P2 time window

The omnibus ANOVA with Sensory Modality and Task revealed a significant main effect of Sensory Modality, $F(2, 58) = 64.40$, $p < .001$, $\eta_p^2 = .689$, showing larger P2 amplitudes for audiovisual stimuli (M = 8.45 μV, SD = 3.56) compared to auditory (M = 6.34 μV, SD = 2.77; $t(29) = -4.85$, $p < .001$) and visual stimuli (M = 2.48 μV, SD = 1.98; $t(29) = -10.97$, $p < .001$), and larger amplitudes for auditory compared to visual stimuli ($t(29) = 6.36$, $p < .001$). No other effect was significant.

To rule out potential confounds related to the sequence of block presentation, the omnibus ANOVA was repeated including sensory modality order as a between-subject factor. This analysis revealed no significant main effect of sensory modality order, $F(2, 27) = 0.50$, $p = .611$, $\eta p^2 = .036$, and no significant interactions involving Order (all $ps \geq .38$), indicating that the observed P2 effects were not influenced by block order.

Figure 5

Topographic Maps of N1 and P2 Components in Experiment 2



Note. Topographic maps show the N1 and P2 components for each sensory modality. Scalp distributions are shown for the N1 and P2 time windows for self-initiated, externally initiated, and difference wave conditions.

DISCUSSION

Experiment 2 replicated Experiment 1 using different visual stimuli, showing N1 attenuation for self-initiated auditory and audiovisual events, but not for visual events. Attenuation effects did not differ between auditory and audiovisual conditions, and no P2 effects were observed. These results indicate that the absence of visual N1 attenuation is independent of stimulus type (i.e., checkerboard reversals or onset stimuli).

GENERAL DISCUSSION

This study compared unisensory sensory attenuation for visual or auditory sensory events with sensory attenuation for multisensory events. We conducted two EEG studies using self-initiated button presses that triggered auditory, visual, or audiovisual events, and compared these with externally initiated events matched in stimulus properties. Given the diverse literature, reporting N1 attenuation (e.g., Gentsch et al., 2012; Gentsch & Schütz-Bosbach, 2011; Mifsud et al., 2018; Ody et al., 2023; Schafer & Marcus, 1973), N1 enhancement (e.g., Balla et al., 2020; Mifsud et al., 2016), or no N1 effects (e.g., Balla et al., 2024; Csifcsák et al., 2019; Hughes & Waszak, 2011), pattern reversals (Exp. 1) and Gabor patch onsets (Exp. 2) were used as visual stimuli. Across both experiments, we observed reliable N1 attenuation (and partial P2 attenuation) for self-initiated auditory and audiovisual stimuli, but not for visual stimuli. N1 responses to self-initiated events were largest for audiovisual events, but comparable effects during externally initiated events suggest this reflects general multisensory integration rather than self-initiation, consistent with prior reports (Giard & Peronnet, 1999; Fleming et al., 2020), demonstrating that audiovisual inputs modulate early sensory processing through multisensory integration mechanisms. ~~general multisensory integration rather than self-initiation, consistent with prior reports (Giard & Peronnet, 1999).~~ Furthermore, N1 attenuation (i.e., the self- vs. external difference) did not differ between unisensory auditory and multisensory audiovisual events, underlining auditory dominance in driving SA.

Self-initiated auditory and audiovisual sensory attenuation

Attenuated N1 responses were consistently found for self-initiated auditory stimuli, in line with prior research (e.g., Baess et al., 2011; Knolle et al., 2013a; Loehr, 2013; van Elk et al., 2014; Timm et al., 2013, 2014). Importantly, our results also showed N1 attenuation for self-initiated audiovisual stimuli, extending previous findings from unisensory to multisensory contexts. The robust N1 attenuation following self-initiated button presses aligns with earlier studies using various approaches and modalities (e.g., ~~Mifsud & Whitford, 2017;~~ Baess et al., 2011; Curio et al., 2000; Ford et al., 2007; Horváth et al., 2012;
Neszmélyi & Horváth, 2021; Numminen & Curio, 1999; Saupe et al., 2013;
Schafer & Marcus, 1973; Sowman et al., 2012~~Neszmélyi & Horváth, 2021; Saupe et al., 2013; Curio et al., 2000; Numminen & Curio, 1999; Baess et al., 2011; Ford et al., 2007; Schafer & Marcus, 1973; Sowman et al., 2012; Horváth et al., 2012~~). Notably, we observed N1 attenuation for multisensory audiovisual stimuli, regardless of delivery method. Furthermore, there was no difference in attenuation effects between audiovisual and auditory modalities. This shows that sensory attenuation operates in a comparable fashion for unisensory and multisensory (audiovisual) stimuli and that the difference between external and self-initiated does not differ across unisensory and multisensory modalities. Thus, the precision in prediction seems to be equivalent

for unisensory and multisensory events. This pattern has important theoretical implications. Predictive processing accounts suggest that combining sensory predictions could enhance precision and thus potentially lead to greater attenuation (Friston, 2005, 2010; Talsma, 2015), but we did not observe such an effect. Instead, these findings are more consistent with internal model frameworks (Wolpert & Kawato, 1998), which emphasize parallel modality-specific, action-dependent predictions, here apparently dominated by the auditory modality. While such a framework may better explain SA for body-related outcomes but not necessarily increase attenuation in vision or audition when action–outcome associations are weak (Dogge et al., 2019). Although both frameworks predict attenuation with reliable sensory predictions, and the observed N1 attenuation in both auditory and audiovisual conditions fit this, neither explains why increased attenuation is not observed when predictions from multiple modalities are combined. Our finding of similar attenuation levels in unisensory and multisensory conditions thus highlights a limitation of current theoretical accounts.

Overall, our findings are consistent with behavioral studies showing multisensory attenuation effects (van Kemenade et al., 2016), suggesting that the brain generates multisensory predictions to anticipate self-generated audiovisual outcomes. Moreover, functional neuroimaging evidence has revealed reduced activity in auditory and visual cortices for self-generated audiovisual stimuli (Straube et al., 2017), further supporting the idea that predictive mechanisms operate across modalities. P2 attenuation appeared only in Experiment 1, mirroring the inconsistent P2 findings in previous studies (e.g. Bolt & Loehr, 2021; Han et al., 2021; Klaffehn et al., 2019~~Han et al., 2021; Klaffehn et al., 2019; Bolt & Loehr, 2021~~). As early ERP components are relatively unaffected by manipulations of attention (Harrison et al., 2021; Timm et al., 2013), the observed P2 reduction is unlikely due to this factor. Its functional significance remains unclear, but recent findings suggest P2 may reflect agency and contextual influences rather than direct predictive processes (Seidel et al., 2021, 2023).

Self-initiated visual sensory attenuation

In contrast to some reports (Mifsud et al., 2016; Ody et al., 2023; Schafer & Marcus, 1973), we found no modulation of N1 or P2 amplitudes for self-initiated visual events, regardless of visual stimulus type. This contradicts prior findings of early visual ERP attenuation for self-initiated stimuli, such as reduced N1 for flashes (Schafer & Marcus, 1973), arrows (Gentsch & Schütz-Bosbach, 2011; Gentsch et al., 2012), a significant reduction in anterior visual N1 amplitude for saccade-initiated flashes, while button-press-initiated flashes showed an intermediate level of attenuation (Mifsud et al., 2018), as well as attenuation for discs (Ody et al., 2023). Other studies report inconsistencies: Balla et al. (2020) found N1 enhancement for self-initiated hand images. Hughes and Waszak (2011) found that self-initiated checkerboard stimuli did not reduce the amplitude of the early visual N1 component, although reductions were observed in later ERP components occurring between 150 and 350 ms after stimulus onset. Similarly, Csifcsák et al. (2019) and Balla et al. (2024) did not find significant N1 suppression for self-initiated hand images or checkerboards. Mifsud et al. (2016) observed enhanced occipital N145 amplitudes, but we did not detect ERP modulation in any early visual component. However, consistent with their findings at fronto-central electrodes, we also found no reliable N1 attenuation for button-press visual effects, regardless of stimulus type. These results highlight substantial variation in visual N1 modulation for self-initiated stimuli across studies and experimental conditions. Nonetheless, our lack of visual attenuation persisted across both pattern reversals and onset presentations, even though these delivery methods corresponded closely with earlier studies.

Some differences in earlier findings may be due to methodological choices. Ody et al. (2023) reported robust N1 and P2 suppression, but their control blocks included a stimulus 1000 ms after the button press, while ours had no post-action sensory event. This suggests visual SA depends on experimental parameters and is less robust than in the auditory modality. Our findings also align with studies reporting no early visual attenuation (Balla et al., 2024; Csifcsák et al., 2019; Hughes & Waszak, 2011) and no behavioral evidence for attenuation (Schwarz et al., 2018). Variability across studies may stem from differences in stimuli, task structure, expectations, and action-stimulus predictability, all of which can influence ERP outcomes (Mifsud et al., 2018).

In addition, the lack of a strong action–outcome association between the button press and visual events in our task may explain the absence of attenuation, with previous research indicating that more reliable and predictable links are necessary to induce attenuation (Mifsud et al., 2018). As such, the lack of anticipatory processing in our paradigm could explain the absence of visual attenuation. A potential limitation of the present studies relates to differences in the temporal characteristics of the stimuli, as the auditory stimuli were shorter in duration than the visual stimuli. Yet, temporally different auditory and visual stimuli have been employed in previous audiovisual studies (e.g., Paris & Davis, 2017; Stekelenburg & Vroomen, 2007; Vroomen & Stekelenburg, 2010), where visual information typically unfolds over longer time scales than auditory events. Such temporal discrepancies could, in principle, influence ERP amplitudes, for example, by introducing sustained neural activity that may overlap with later ERP components. However, the latency and amplitude of early ERP components such as the N1 and P2 appear relatively robust to variations in stimulus duration, whereas duration-related effects are evident for later components (Polich, 1989). Taken together, differences in stimulus duration may have influenced the overall temporal profile of the ERP, particularly for later components, but are unlikely to account for the modality-specific effects observed in the N1and P2 time windows.

Conclusion

As everyday perception often relies on multisensory integration, it is essential to investigate sensory attenuation as an index of self–other discrimination. The present findings extend previous unisensory research by demonstrating reliable N1 sensory attenuation in response to simultaneous auditory and visual stimuli across two experiments. Notably, in the present paradigm, audiovisual outcomes did not increase attenuation beyond that observed for auditory outcomes. While these results suggest the existence of a parallel, action-dependent mechanism of sensory attenuation operating across modalities, the current design does not allow for a clear distinction between internal model and predictive processing accounts.

Overall, the results emphasize the importance of considering multisensory stimulation in the modulation of early sensory processing and highlight the need for future research to more directly test how predictions across modalities are combined and how different theoretical accounts can be empirically distinguished.~~Since everyday perception often relies on multisensory integration, investigating sensory attenuation as an index of self-other discrimination is essential. Our results extend prior~~

~~unisensory research by showing robust N1 sensory attenuation across two experiments in response to simultaneous auditory and visual stimuli, with the auditory modality predominantly driving this effect. Overall, our findings suggest that both internal prediction mechanisms, through the assumption of multiple inverse and forward models and the predictive processing framework, which assumes that the brain continuously generates and updates predictions about multisensory input to minimize prediction errors, provide valuable perspectives for understanding sensory attenuation in multisensory contexts. However, both frameworks fall short in explaining how different predictions are integrated. These findings pave the way for future research on the complex dynamics of sensory processing and how interactions between modalities shape perception.~~

APPENDIX

This appendix reports additional analyses conducted to further examine the contribution of visual and multisensory processes to the observed effects. First, for each experiment analyses of visual components (P1 and N2) at the occipital electrode Oz are reported to account for the modality-specific topography of visual evoked responses. Second, a posterior region-of-interest (ROI) analysis including electrodes O1, O2, Oz, PO3, and PO4 was conducted to assess whether the observed effects generalize across parieto-occipital electrodes. Third, an additivity analysis was performed to evaluate multisensory integration by comparing the summed unisensory auditory and visual responses (A+V) with the audiovisual responses (AV). Finally, control analyses at Cz using alternative time windows are reported.

Experiment 1

1. Oz electrode analyses

To account for modality-specific topography of visual evoked responses, additional analyses were conducted at the Oz electrode. The time windows for the visual P1 and N2 components were determined by inspecting the grand-average waveforms across all visual and audiovisual conditions. Peak latencies were identified from these waveforms, and time windows were defined around these peaks. The resulting time windows were 106-146 ms for the P1 component and 226-266 ms for the N2 component.

P1 time window

A repeated-measures ANOVA with the factors Sensory Modality and Task revealed a significant main effect of Sensory Modality, $F(1, 22) = 9.21$, $p = .006$, $\eta p^2 = .295$, indicating larger P1 amplitudes for audiovisual ($M = 8.04$ μV, $SD = 3.06$) compared to visual stimuli ($M = 5.87$ μV, $SD = 3.68$). There was no main effect of Task, $F(1, 22) =$

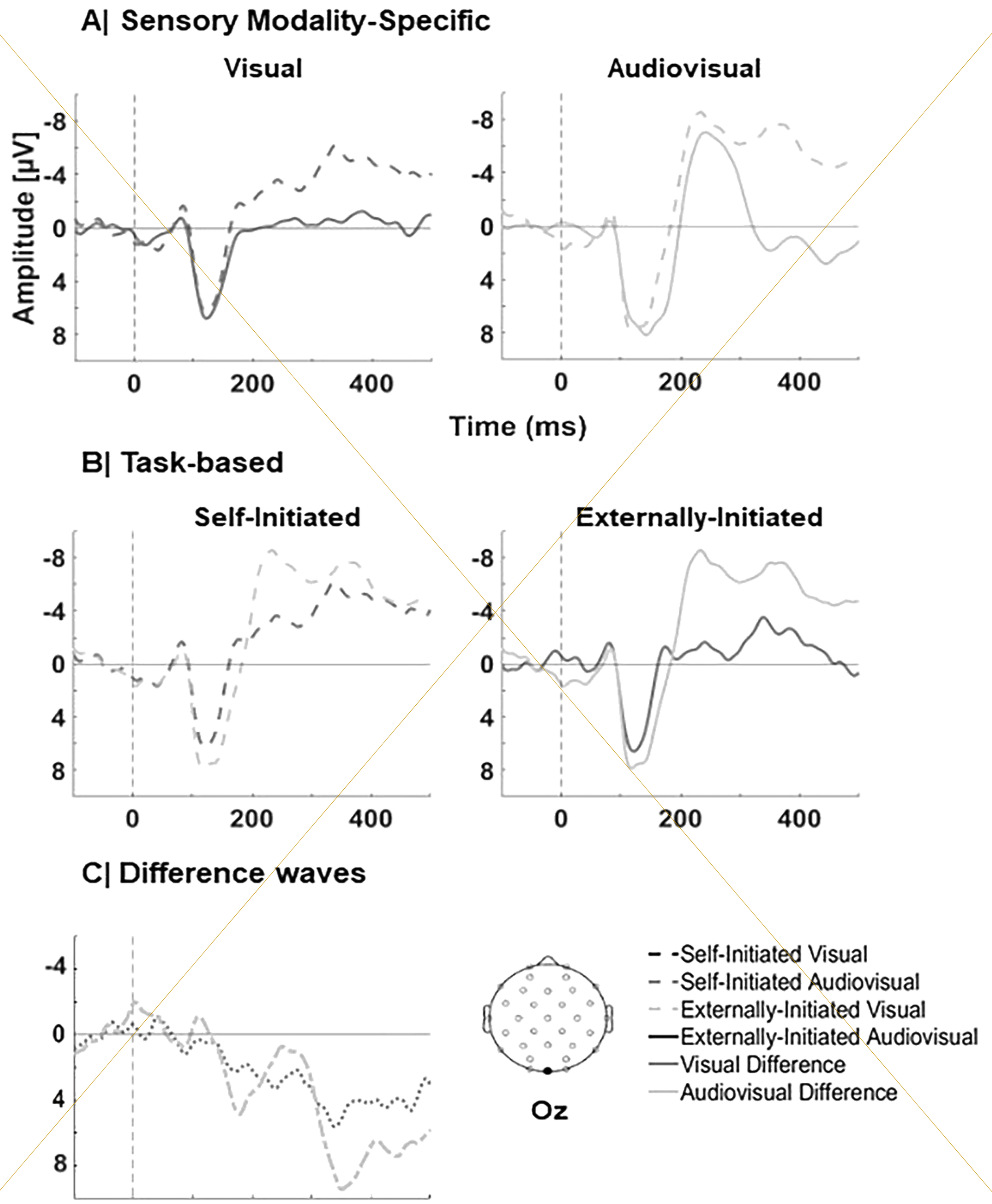
0.03, $p = .866$, $\eta p^2 = .001$. Likewise, the interaction between Sensory Modality and Task was not significant, $F(1, 22) = 0.58$, $p = .454$, $\eta p^2 = .026$.

N2 time window

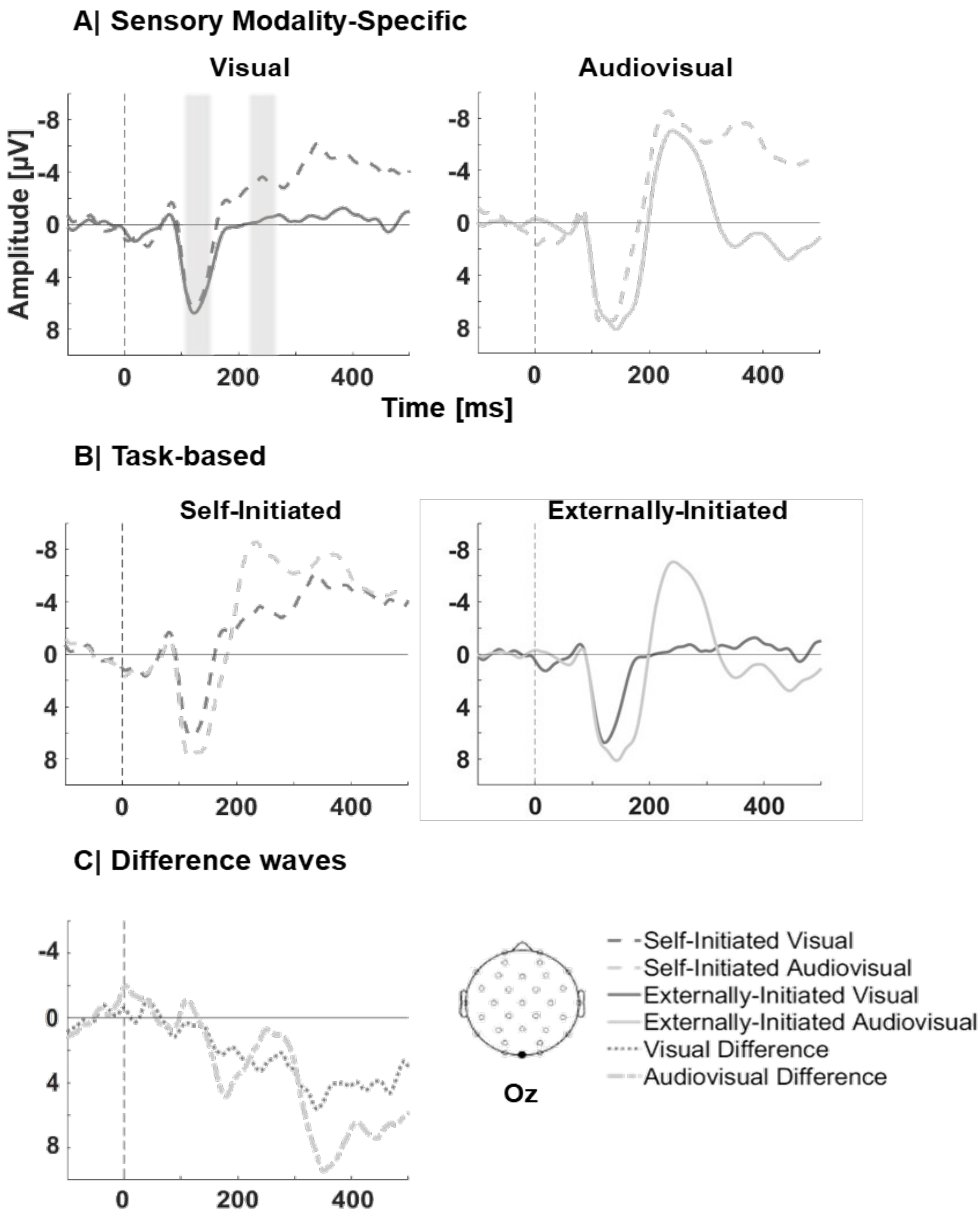
A repeated-measures ANOVA with the factors Sensory Modality and Task revealed a significant main effect of Sensory Modality, $F(1, 22) = 50.34$, $p < .001$, $\eta p^2 = .696$, indicating larger N2 amplitudes for audiovisual (M = -8.58 µV, SD = 4.60) compared to visual stimuli (M = -2.45 µV, SD = 3.11). There was no main effect of Task, $F(1, 22) = 4.07$, $p = .056$, $\eta p^2 = .156$. Likewise, the interaction between Sensory Modality and Task was not significant, $F(1, 22) = 1.52$, $p = .230$, $\eta p^2 = .065$.

Figure A1

Grand-average ERPs for visual and audiovisual conditions (Experiment 1)~~*Grand Average ERPs (Fz-referenced) for Each Sensory Modality in Self-Initiated and*~~



~~*Externally initiated Conditions (Experiment 1)*~~



Note. Grand average waveforms (Fz-referenced) at the Oz electrode in Experiment 1 for (A) Visual, and audiovisual modalities, plotted separately for self-initiated (dashed lines) and externally initiated (solid lines) stimuli. (B) Self-initiated and externally initiated tasks are displayed for each sensory modality, with self-initiated conditions represented by dashed lines and externally initiated conditions by solid lines. (C) Difference waves, computed as externally minus self-initiated conditions, are shown for each modality.

2. Posterior ROI analysis

P1 time window

A repeated-measures ANOVA with the factors Sensory Modality and Task revealed a significant main effect of Sensory Modality, $F(1, 22) = 9.10$, $p = .006$, $\eta_p^2 = .293$, indicating larger P1 amplitudes for audiovisual ($M = 7.39$ μV, $SD = 2.85$) compared to visual stimuli ($M = 5.52$ μV, $SD = 3.00$). There was no main effect of Task, $F(1, 22) = 0.005$, $p = .945$, $\eta_p^2 = .0002$. Likewise, the interaction between Sensory Modality and Task was not significant, $F(1, 22) = 0.25$, $p = .620$, $\eta_p^2 = .011$.

N2 time window

A repeated-measures ANOVA with the factors Sensory Modality and Task revealed a significant main effect of Sensory Modality, $F(1, 22) = 44.23$, $p < .001$, $\eta_p^2 = .668$, indicating larger N2 amplitudes for audiovisual ($M = -7.27$ μV, $SD = 4.46$) compared to visual stimuli ($M = -1.83$ μV, $SD = 3.04$). There was no main effect of Task, $F(1, 22) = 4.16$, $p = .053$, $\eta_p^2 = .159$. Likewise, the interaction between Sensory Modality and Task was not significant, $F(1, 22) = 1.14$, $p = .298$, $\eta_p^2 = .049$.

3. Additivity analysis

4. Alternative time-window analyses

N1 time window

A repeated-measures ANOVA with the factors Sensory Modality and Task revealed a significant main effect of Sensory Modality, $F(2, 44) = 23.82$, $p < .001$, $\eta p^2 = .520$, indicating larger N1 amplitudes for audiovisual (M = -4.52 μV, SD = 2.43) compared to auditory stimuli (M = -3.62 μV, SD = 2.01), $t(22) = 4.12$, $p < .001$, and visual stimuli (M = -1.46 μV, SD = 1.63), $t(22) = 8.00$, $p < .001$. Additionally, auditory stimuli elicited significantly larger N1 amplitudes than visual stimuli, $t(22) = -4.52$, $p < .001$. A significant main effect of Task was also observed, $F(1, 22) = 4.58$, $p = .044$, $\eta p^2 = .172$, with larger N1 amplitudes for externally initiated stimuli (M = -3.22 μV, SD = 1.76) compared to self-initiated stimuli (M = -2.76 μV, SD = 2.08). Importantly, the interaction between Sensory Modality and Task was significant, $F(2, 44) = 4.39$, $p = .018$, $\eta p^2 = .166$.

Post hoc t-tests showed significantly reduced N1 amplitudes for self-initiated (M = -3.05 μV, SD = 2.16) compared to externally initiated stimuli (M = -4.20 μV, SD = 2.35) in the auditory condition, $t(22) = 2.68$, $p = .014$, and for self-initiated (M = -3.97 μV, SD = 2.54) compared to externally initiated stimuli (M = -5.08 μV, SD = 2.71) in the audiovisual condition, $t(22) = 2.70$, $p = .013$. In contrast, no significant difference was observed for self-initiated (M = -1.56 μV, SD = 2.26) compared to externally initiated stimuli (M = -1.36 μV, SD = 1.66) in the visual condition, $t(22) = -0.42$, $p = .682$.

Further comparisons between sensory modalities revealed that, for self-initiated stimuli, audiovisual amplitudes (M = -3.97 μV, SD = 2.54) were significantly larger than visual stimuli (M = -1.56 μV, SD = 2.26), $t(22) = 4.83$, $p < .001$, and auditory stimuli (M = -3.05 μV, SD = 2.16) were significantly larger than visual stimuli, $t(22) = -2.65$, $p = .015$, whereas auditory and audiovisual stimuli did not differ, $t(22) = 1.84$, $p = .079$. For externally initiated stimuli, audiovisual responses (M = -5.08 μV, SD = 2.71) were significantly larger than visual stimuli (M = -1.36 μV, SD = 1.66), $t(22) = 6.39$, $p < .001$, and auditory stimuli elicited significantly larger N1 amplitudes than visual stimuli, $t(22) = -4.98$, $p < .001$. In contrast, audiovisual and auditory stimuli (M = -4.20 μV, SD = 2.35) did not differ, $t(22) = 2.13$, $p = .044$.

The attenuation effect did not differ between the auditory and visual modalities, $t(22) = -1.74$, $p = .096$, nor between the audiovisual and auditory modalities, $t(22) = 0.38$, $p = .706$, or between the audiovisual and visual modalities, $t(22) = 2.26$, $p = .034$.

P2 time window

A repeated-measures ANOVA with the factors Sensory Modality and Task revealed a significant main effect of Sensory Modality, $F(1.51, 33.31) = 65.18$, $p < .001$, $\eta p^2 = .748$, indicating larger P2 amplitudes for audiovisual (M = 9.36 μV, SD = 3.83) compared to auditory stimuli (M = 6.53 μV, SD = 2.86), $t(22) = -7.38$, $p < .001$, and visual stimuli (M = 2.86 μV, SD = 2.33), $t(22) = -9.64$, $p < .001$. Additionally, auditory stimuli elicited significantly larger P2 amplitudes than visual stimuli, $t(22) = 5.99$, $p < .001$. A significant main effect of Task was also observed, $F(1, 22) = 17.10$, $p < .001$, $\eta p^2 = .437$, with larger P2 amplitudes for externally initiated stimuli (M = 7.09 μV, SD = 3.31) compared to self-initiated stimuli (M = 5.14 μV, SD = 2.71). Importantly, the interaction between Sensory Modality and Task was significant, $F(1.48, 32.63) = 13.76$, $p < .001$, $\eta p^2 = .385$.

Post hoc t-tests showed significantly reduced P2 amplitudes for self-initiated (*M* = 5.18 μV, *SD* = 2.50) compared to externally initiated stimuli (*M* = 7.89 μV, *SD* = 3.82) in the auditory condition, $t(22) = -4.34$, $p < .001$, and for self-initiated (*M* = 7.41 μV, *SD* = 3.38) compared to externally initiated stimuli (*M* = 11.31 μV, *SD* = 5.08) in the audiovisual condition, $t(22) = -4.75$, $p < .001$. In contrast, no significant difference was observed for self-initiated (*M* = 2.86 μV, *SD* = 3.13) compared to externally initiated stimuli (*M* = 2.87 μV, *SD* = 2.30) in the visual condition, $t(22) = -0.01$, $p = .989$.

Further comparisons between sensory modalities revealed that, for self-initiated stimuli, audiovisual amplitudes (*M* = 7.41 μV, *SD* = 3.38) were significantly larger than both auditory, $t(22) = -4.52$, $p < .001$, and visual stimuli, $t(22) = -6.08$, $p < .001$. Moreover, auditory amplitudes (*M* = 5.18 μV, *SD* = 2.50) were significantly larger than visual stimuli (*M* = 2.86 μV, *SD* = 3.13), $t(22) = 3.28$, $p = .003$. For externally initiated stimuli, audiovisual responses (*M* = 11.31 μV, *SD* = 5.08) were significantly larger than both auditory (*M* = 7.89 μV, *SD* = 3.82), $t(22) = -7.60$, $p < .001$, and visual stimuli (*M* = 2.87 μV, *SD* = 2.30), $t(22) = -9.48$, $p < .001$. Additionally, auditory stimuli elicited larger P2 amplitudes than visual stimuli, $t(22) = 6.98$, $p < .001$.

To further examine whether P2 attenuation differed across sensory modalities, difference waves were compared. The attenuation effect was larger for the auditory modality (*M* = 3.50 μV, *SD* = 3.27) than for the visual modality (*M* = 0.56 μV, *SD* = 3.16), $t(22) = 3.92$, $p < .001$. The attenuation effect for the audiovisual modality (*M* = 4.21 μV, *SD* = 3.96) was larger than for the visual modality, $t(22) = -4.05$, $p < .001$.

In contrast, the attenuation effect for the audiovisual modality did not differ from the auditory modality, $t(22) = -1.37$, $p = .184$.

Experiment 2

1. Oz electrode analyses

To account for modality-specific topography of visual evoked responses, additional analyses were conducted at the Oz electrode. The time windows for the visual P1 and N2 components were determined by inspecting the grand-average waveforms across all visual and audiovisual conditions. Peak latencies were identified from these waveforms, and time windows were defined around these peaks. The resulting time windows were 122–162 ms for the P1 component and 202–242 ms for the N2 component.

P1 time window

A repeated-measures ANOVA with the factors Sensory Modality and Task revealed a significant main effect of Sensory Modality, $F(1, 29) = 6.71$, $p = .015$, $\eta p^2 = .188$, indicating larger P1 amplitudes for audiovisual ($M = 7.85$ µV, $SD = 3.95$) compared to visual stimuli ($M = 6.12$ µV, $SD = 3.03$). A significant main effect of Task was observed, $F(1, 29) = 40.80$, $p < .001$, $\eta p^2 = .585$, with larger amplitudes for externally initiated stimuli ($M = 8.31$ µV, $SD = 3.15$) compared to self-initiated stimuli ($M = 5.65$ µV, $SD = 3.29$). Importantly, the interaction between Sensory Modality and Task was significant, $F(1, 29) = 6.56$, $p = .016$, $\eta p^2 = .184$. Post hoc t-tests showed significantly reduced P1 amplitudes for self-initiated compared to externally initiated stimuli in both the visual condition (self: $M = 5.19$ µV, $SD = 3.65$; external: $M = 7.05$ µV, $SD = 3.04$),

$t(29) = -3.50$, $p = .002$, and the audiovisual condition (self: $M = 6.12$ μV, $SD = 4.18$; external: $M = 9.57$ μV, $SD = 4.21$), $t(29) = -6.82$, $p < .001$.

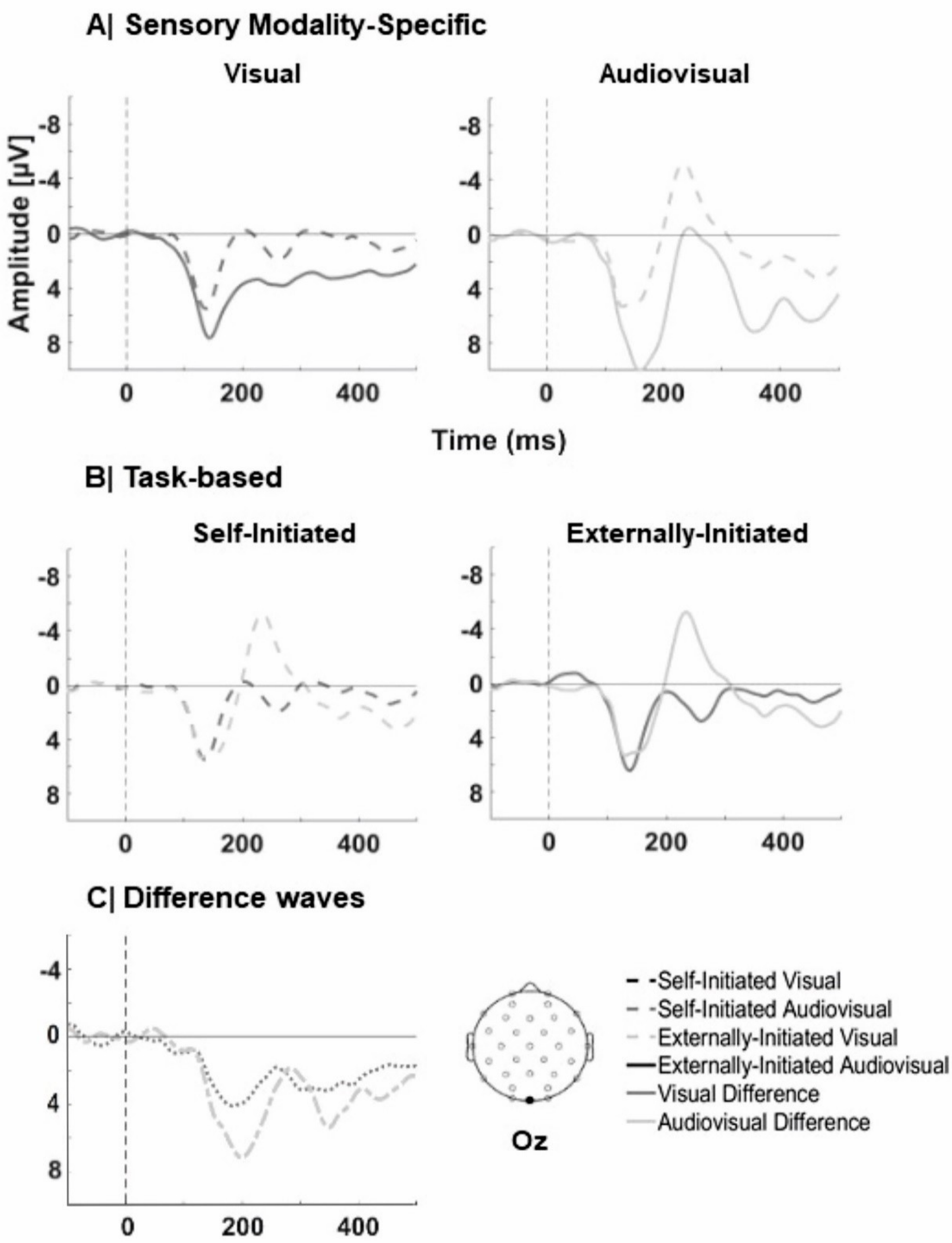
Further comparisons between sensory modalities revealed no difference between visual and audiovisual conditions for self-initiated stimuli (visual: $M = 5.19$ μV, $SD = 3.65$; audiovisual: $M = 6.12$ μV, $SD = 4.18$), $t(29) = -1.19$, $p = .244$, whereas externally initiated audiovisual stimuli ($M = 9.57$ μV, $SD = 4.21$) elicited significantly larger amplitudes than visual stimuli ($M = 7.05$ μV, $SD = 3.04$), $t(29) = -3.67$, $p < .001$.

N2 time window

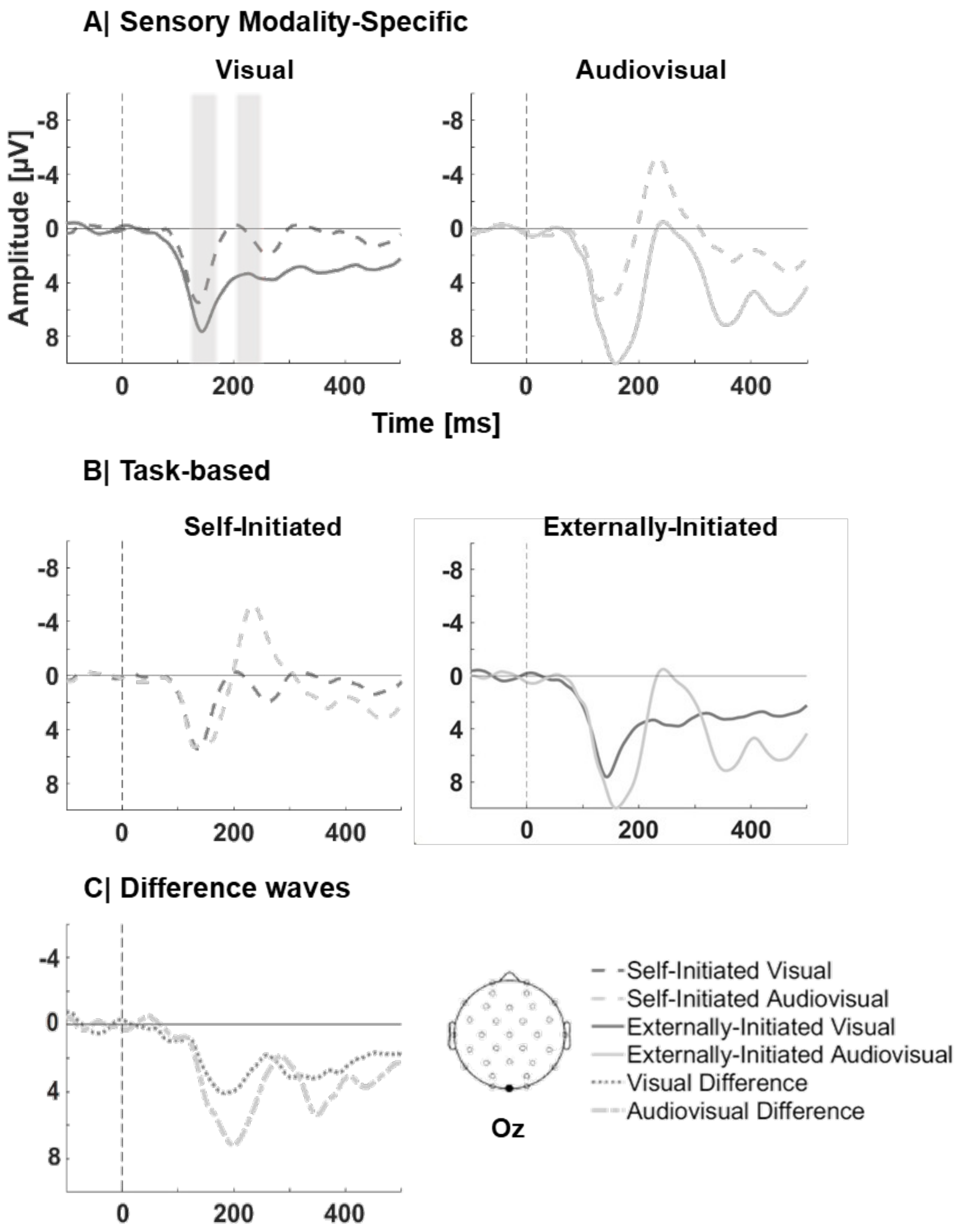
A repeated-measures ANOVA with the factors Sensory Modality and Task revealed a significant main effect of Sensory Modality, $F(1, 29) = 16.69$, $p < .001$, $\eta p^2 = .365$, indicating larger N2 amplitudes for audiovisual ($M = -2.78$ μV, $SD = 6.02$) compared to visual stimuli ($M = 0.85$ μV, $SD = 4.17$). A significant main effect of Task was observed, $F(1, 29) = 27.81$, $p < .001$, $\eta p^2 = .490$, with larger amplitudes for self-initiated stimuli ($M = -3.19$ μV, $SD = 6.15$) compared to externally initiated stimuli ($M = 1.27$ μV, $SD = 3.82$). The interaction between Sensory Modality and Task was not significant, $F(1, 29) = 0.86$, $p = .362$, $\eta p^2 = .029$.

Figure A2

Grand-average ERPs for visual and audiovisual conditions (Experiment 2)~~*Grand*~~



~~*Average ERPs (Fz-referenced) for Each Sensory Modality in Self-Initiated and Externally initiated Conditions (Experiment 2)*~~



Note. Grand average waveforms (Fz-referenced) at the Oz electrode in Experiment 2 for (A) Visual, and audiovisual modalities, plotted separately for self-initiated (dashed lines) and externally initiated (solid lines) stimuli. (B) Self-initiated and externally initiated tasks are displayed for each sensory modality, with self-initiated conditions represented by dashed lines and externally initiated conditions by solid lines. (C) Difference waves, computed as externally minus self-initiated conditions, are shown for each modality.

2. Posterior ROI analysis

P1 time window

A repeated-measures ANOVA with the factors Sensory Modality and Task revealed a significant main effect of Sensory Modality, $F(1, 29) = 5.57$, $p = .025$, $\eta p^2 = .161$, indicating larger P1 amplitudes for audiovisual ($M = 7.13$ µV, $SD = 3.88$) compared to visual stimuli ($M = 5.63$ µV, $SD = 2.75$). A significant main effect of Task was also observed, $F(1, 29) = 40.54$, $p < .001$, $\eta p^2 = .583$, with larger amplitudes for externally initiated stimuli ($M = 7.71$ µV, $SD = 2.96$) compared to self-initiated stimuli ($M = 5.06$ µV, $SD = 3.23$). Importantly, the interaction between Sensory Modality and Task was significant, $F(1, 29) = 4.81$, $p = .036$, $\eta p^2 = .142$.

Post hoc t-tests showed significantly reduced P1 amplitudes for self-initiated compared to externally initiated stimuli in both the visual condition (self: $M = 4.65$ µV, $SD = 3.42$; external: $M = 6.62$ µV, $SD = 2.80$), $t(29) = -3.64$, $p = .001$, and the audiovisual condition (self: $M = 5.46$ µV, $SD = 4.18$; external: $M = 8.80$ µV, $SD = 4.06$), $t(29) = -6.67$, $p < .001$. Further comparisons between sensory modalities revealed no significant difference between visual and audiovisual conditions for self-initiated stimuli (visual: $M = 4.65$ µV, $SD = 3.42$; audiovisual: $M = 5.46$ µV, $SD = 4.18$), $t(29) = -1.10$, $p = .281$, whereas externally initiated audiovisual stimuli ($M = 8.80$ µV, $SD = 4.06$) elicited significantly larger amplitudes than externally initiated visual stimuli ($M = 6.62$ µV, $SD = 2.80$), $t(29) = -3.25$, $p = .003$.

N2 time window

A repeated-measures ANOVA with the factors Sensory Modality and Task revealed a significant main effect of Sensory Modality, $F(1, 29) = 13.30$, $p = .001$, $\eta p^2 = .314$, indicating larger N2 amplitudes for audiovisual ($M = -1.78$ µV, $SD = 5.57$) compared to visual stimuli (M = 1.29 μV, SD = 4.39). A significant main effect of Task was also observed, $F(1, 29) = 33.31$, $p < .001$, $\eta p^2 = .535$, with larger amplitudes for self-initiated stimuli (M = −2.43 μV, SD = 5.82) compared to externally initiated stimuli (M = 1.94 μV, SD = 3.78). The interaction between Sensory Modality and Task was not significant, $F(1, 29) = 2.21$, $p = .148$, $\eta p^2 = .071$.

3. Additivity analysis

4. Alternative time-window analyses

N1 time window

A repeated-measures ANOVA with the factors Sensory Modality and Task revealed a significant main effect of Sensory Modality, $F(2, 58) = 35.61$, $p < .001$, $\eta p^2 = .551$, indicating larger N1 amplitudes for audiovisual (M = -4.55 μV, SD = 2.34) compared to auditory stimuli (M = -3.40 μV, SD = 2.02), $t(29) = 4.12$, $p < .001$, and visual stimuli (M = -1.62 μV, SD = 1.22), $t(29) = 8.00$, $p < .001$. Additionally, auditory stimuli elicited significantly larger N1 amplitudes than visual stimuli, $t(29) = -4.52$, $p < .001$. A significant main effect of Task was also observed, $F(1, 29) = 10.07$, $p = .004$, $\eta p^2 = .258$, with larger N1 amplitudes for externally initiated stimuli (M = -3.35 μV, SD = 1.64) compared to self-initiated stimuli (M = -2.82 μV, SD = 1.82). Importantly, the interaction between Sensory Modality and Task was significant, $F(2, 58) = 9.69$, $p < .001$, $\eta p^2 = .250$.

Post hoc t-tests showed significantly reduced N1 amplitudes for self-initiated (*M* = -2.78 μV, *SD* = 2.53) compared to externally initiated stimuli (*M* = -4.01 μV, *SD* = 1.84) in the auditory condition, $t(29) = 3.79$, $p < .001$, and for self-initiated (*M* = -3.90 μV, *SD* = 2.45) compared to externally initiated stimuli (*M* = -5.20 μV, *SD* = 2.63) in the audiovisual condition, $t(29) = 3.56$, $p = .001$. In contrast, no significant difference was observed for self-initiated (*M* = -1.73 μV, *SD* = 1.82) compared to externally initiated stimuli (*M* = -1.50 μV, *SD* = 0.99) in the visual condition, $t(29) = -0.75$, $p = .461$. Further comparisons between sensory modalities revealed that, for self-initiated stimuli, audiovisual amplitudes (*M* = -3.90 μV, *SD* = 2.45) were significantly larger than both auditory (*M* = -2.78 μV, *SD* = 2.53), $t(29) = 3.24$, $p = .003$, and visual stimuli (*M* = -1.73 μV, *SD* = 1.82), $t(29) = 5.15$, $p < .001$, whereas auditory and visual stimuli did not differ, $t(29) = -1.94$, $p = .062$. For externally initiated stimuli, audiovisual responses (*M* = -5.20 μV, *SD* = 2.63) were significantly larger than both auditory (*M* = -4.01 μV, *SD* = 1.84), $t(29) = 3.63$, $p = .001$, and visual stimuli (*M* = -1.50 μV, *SD* = 0.99), $t(29) = 9.02$, $p < .001$. Additionally, auditory stimuli elicited larger N1 amplitudes than visual stimuli, $t(29) = -8.05$, $p < .001$.

To further examine whether N1 attenuation differed across sensory modalities, difference waves (externally initiated minus self-initiated) were compared. The attenuation effect was larger for the auditory modality (*M* = -1.81 μV, *SD* = 2.36) than for the visual modality (*M* = -0.75 μV, *SD* = 2.08), $t(29) = -2.06$, $p = .049$. In contrast, the attenuation effect for the audiovisual modality (*M* = -1.47 μV, *SD* = 2.15) did not differ from the auditory modality, $t(29) = -0.61$, $p = .546$, nor from the visual modality, $t(29) = 1.59$, $p = .123$.

P2 time window

A repeated-measures ANOVA with the factors Sensory Modality and Task revealed a significant main effect of Sensory Modality, $F(2, 58) = 70.78$, $p < .001$, $\eta p^2 = .709$, indicating larger P2 amplitudes for audiovisual ($M = 8.33$ µV, $SD = 3.49$) compared to auditory stimuli ($M = 6.26$ µV, $SD = 2.81$), $t(29) = -5.23$, $p < .001$, and visual stimuli ($M = 2.31$ µV, $SD = 1.93$), $t(29) = -11.29$, $p < .001$. Additionally, auditory stimuli elicited significantly larger P2 amplitudes than visual stimuli, $t(29) = 6.66$, $p < .001$. There was no significant main effect of Task, $F(1, 29) = 1.51$, $p = .230$, $\eta p^2 = .049$. Likewise, the interaction between Sensory Modality and Task was not significant, $F(2, 58) = 0.98$, $p = .380$, $\eta p^2 = .033$.

Ethical Statement

No animal studies are presented in this manuscript. All procedures performed in the experiments involving human subjects were in accordance with the ethical standards of the institutional research committees and with the 1964 Declaration of Helsinki and its later amendments or comparable ethical standards. Our subjects were ~~written~~ informed about the experiment and their rights. They provided written informed consent. All experiments were approved by the Local Ethical Committee of Faculty 1 at the University of Hildesheim (Approval number 225). No potentially identifiable human images or data are presented in the manuscript.

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Data Availability Statement

The data used for the statistical analysis presented in this article are available on OSF here: https://osf.io/7h9by/?view_only=d4708574e9894fd692e4a2d9d94cb09e

Author Contributions

Shabnamalsadat Ayatollahi: Data curation; Formal analysis; Visualization; Writing — Original draft; Writing — Review & editing. Christina Bermeitinger: Supervision; Writing — Review & editing. Pamela Baess: Conceptualization; Methodology; Software; Investigation; Supervision; Writing — Original draft; Writing — Review & editing.

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